Project Report: Online Retail Product Recommendation System

1. Project Objective

In this project, I built a product recommendation system using customer purchase data. My goal was to analyze customers' previous purchases and recommend three products to each customer.

2. Data Cleaning and Preparation

First, I loaded the dataset using Python and Pandas and checked its structure and contents. I removed duplicate records and rows with missing important information, such as customer IDs or invoice dates. I also removed cancelled invoices and records with zero or negative quantities or prices.

Next, I added a column called Revenue to calculate the revenue for each transaction line by multiplying the quantity by the unit price. I also excluded non-product codes, such as C2 and M, from the data used for product recommendations.

3. Exploratory Data Analysis

After cleaning the data, I explored the dataset to understand the customers, products, and transactions. I analyzed the best-selling products, the countries with the highest purchase activity, and monthly revenue trends.

This analysis helped me understand how transaction data could be used to identify customer purchasing patterns and generate relevant product recommendations.

4. Customer Segmentation

To better understand customer behavior, I used RFM analysis. I examined three main factors:

Recency: How recently a customer made a purchase.

Frequency: How often a customer made purchases.

Monetary: How much a customer spent in total.

Based on these factors, I grouped customers into different segments, including Loyal/Active, Regular, Low Engagement, and At Risk.

This segmentation helped me gain a better understanding of customer purchasing behavior.

5. Building the Product Recommendation System

I developed a hybrid recommendation system that combined three approaches:

Repeat purchases: Recommending products that a customer had purchased multiple times.

Recent purchases: Using the customer's recent purchase history to generate recommendations.

Similar customers: Recommending products purchased by customers with similar buying patterns.

To identify similar customers, I used Cosine Similarity to compare their purchase patterns and find customers with similar product preferences.

I also developed an improved version of the system that used popular products to fill missing recommendation slots when necessary. I tried to avoid recommending products that customers had already purchased or that were already included in their recommendation lists.

6. Model Evaluation

To evaluate the recommendation system, I split the data based on time. I used earlier transactions to build the model and the final 30 days of transactions to evaluate its recommendations.

I compared the recommended products with customers' actual purchases during the evaluation period.

I used two main evaluation metrics:

Precision: The proportion of recommended product slots that matched products purchased during the evaluation period.

Customer Success Rate: The percentage of evaluated customers who purchased at least one recommended product during the evaluation period.

I also compared the original hybrid model with an improved version that used popular products to fill missing recommendation slots. This helped me examine how the different approaches performed.

7. Final Results

At the end of the project, I created a recommendation table for 10 sample customers, with three recommended products for each customer. I also prepared a simulated conversation to demonstrate how the recommendation system could present its suggestions to a customer.

The final outputs showed how purchase history and customer similarity could be used to generate personalized product recommendations.

8. Comparison of Approaches and Model Limitations

Throughout the project, I tested several recommendation approaches and compared their results to understand which methods could generate more relevant suggestions.

I explored repeat-purchase recommendations, recent-purchase recommendations, and recommendations based on similar customers. I then combined these approaches and tested an additional version that used popular products when the original model could not provide three recommendations.Although I experimented with different approaches, I did not achieve a very high success rate. The results suggested that the hybrid model had potential, but there was still room for improvement. I also learned that providing three recommendations to every customer does not necessarily mean that all three recommendations are relevant.One limitation of this project is that future customer behavior cannot always be predicted from previous purchases. In future work, I could test additional recommendation methods and compare them with a simple baseline, such as recommending the most popular products, to better understand the model's performance.